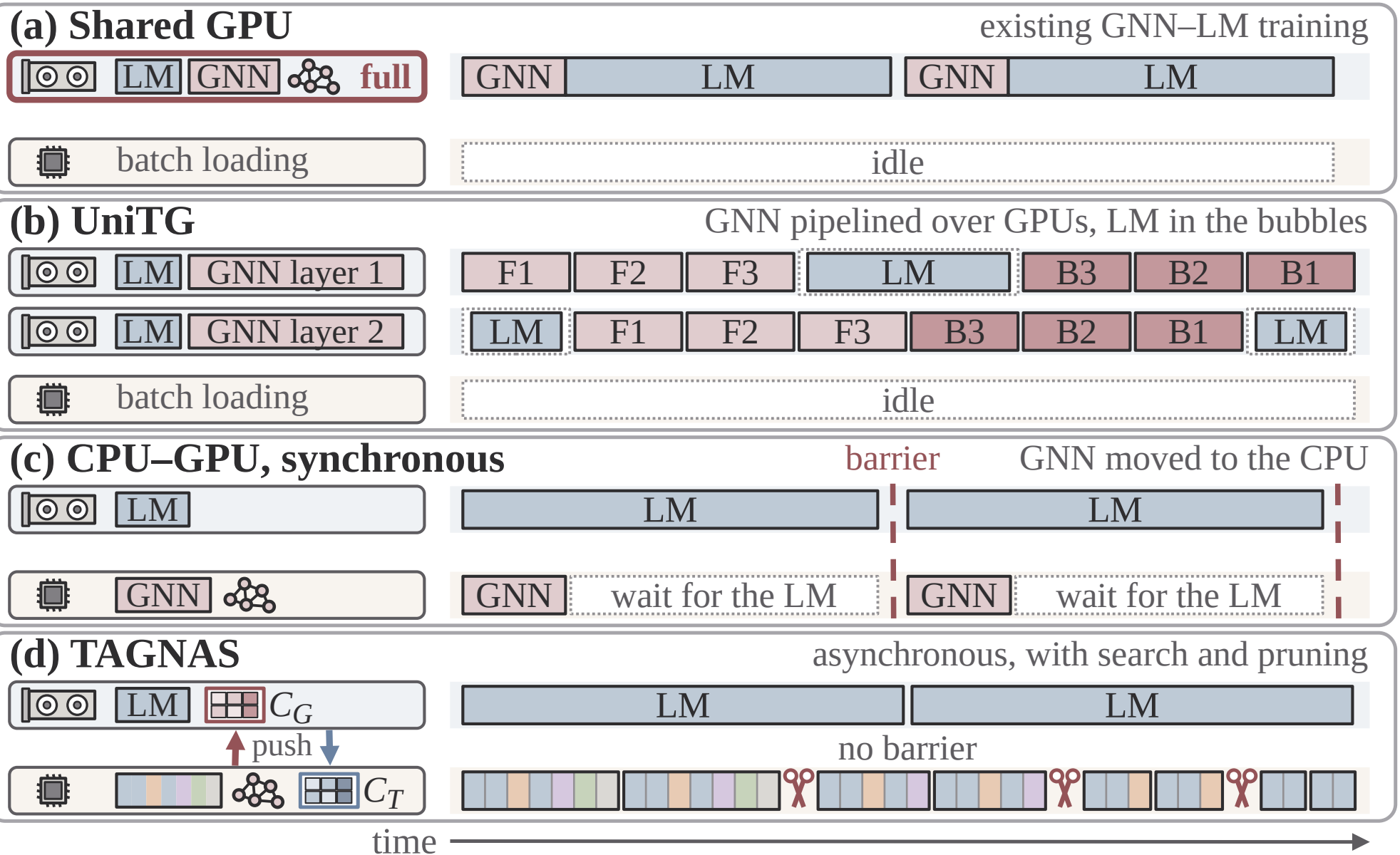
(a) Shared GPU
existing GNN–LM training
LM
GNN
full
GNN
LM
GNN
LM
batch loading
idle
(b) UniTG
GNN pipelined over GPUs, LM in the bubbles
LM
GNN layer 1
F1
F2
F3
LM
B3
B2
B1
LM
GNN layer 2
LM
F1
F2
F3
B3
B2
B1
LM
batch loading
idle
(c) CPU–GPU, synchronous
barrier
GNN moved to the CPU
LM
LM
LM
GNN
GNN
wait for the LM
GNN
wait for the LM
(d) TAGNAS
asynchronous, with search and pruning
LM
C_G
LM
LM
push
no barrier
C_T
time